SOC Detection & Threat Hunting Platform

Project Report

Azeem Abdulla
SOC Analyst (L1) | Cybersecurity Graduate
azeemabdulla67@gmail.com | github.com/azeem7012

May 2026

1. Executive Summary

This report documents the design, development, and findings of an Advanced SOC Detection and Threat Hunting Platform built as a final-year cybersecurity project. The platform is a full-stack Security Operations Center (SOC) dashboard that provides real-time alert triage, MITRE ATT&CK technique mapping, IP risk scoring, and an interactive threat hunt panel.

The project directly extends a custom log-based Intrusion Detection System (IDS) built during an internship at Cloudisian, which detects SSH brute-force attacks (T1110), port scanning (T1046), and web enumeration (T1190) from real auth.log and syslog data. This platform ingests those structured alerts and visualizes them in a SOC analyst workflow.

2. Project Overview

2.1 Objectives

- Build a full-stack SOC dashboard connecting a custom IDS to a real analyst workflow UI
- Visualize MITRE ATT&CK technique coverage using a color-coded heatmap
- Automate IP risk scoring based on alert severity and frequency
- Integrate VirusTotal API for live IOC enrichment on source IPs
- Provide a threat hunt panel for real-time search and filtering of alerts

2.2 Tech Stack

Layer	Technology

Frontend	React, Tailwind CSS, Axios
Backend	Python, Flask, Flask-CORS
Threat Intel	VirusTotal API v3
Detection Data	Custom log-based IDS alert output (JSON)
Framework	MITRE ATT&CK
Version Control	Git, GitHub

2.3 Project Structure

File / Folder	Purpose
backend/app.py	Flask REST API with 5 endpoints
backend/data/alerts.json	IDS alert data (MITRE-mapped JSON)
frontend/src/App.js	Main React dashboard component
frontend/src/components/StatCard.jsx	Severity stat cards
frontend/src/components/AlertTable.jsx	Live alert feed table
frontend/src/components/MitreHeatmap.jsx	MITRE ATT&CK heatmap
frontend/src/components/RiskScorePanel.jsx	IP risk score bars
frontend/src/components/ThreatHuntPanel.jsx	Search/filter panel

3. System Architecture

The platform follows a client-server architecture with a React frontend communicating with a Python Flask backend over REST API calls. The backend reads structured JSON alerts generated by the custom IDS and exposes them through 5 dedicated endpoints.

3.1 API Endpoints

Method	Endpoint	Description
GET	/api/alerts	All alerts with optional severity/technique/status filters
GET	/api/stats	Summary stats — total, critical, high, open, technique counts
GET	/api/incidents	Alerts grouped by source IP for correlation
GET	/api/risk	Automated risk score per source IP
GET	/api/enrich/<ip>	VirusTotal enrichment — verdict, country, owner, detection stats

3.2 Alert Schema

Each alert generated by the IDS follows this JSON structure:

```
{ "id": "ALT-001", "timestamp": "2026-05-13T08:14:32Z", "source_ip": "192.168.1.105",
  "technique_id": "T1110", "technique_name": "Brute Force",
  "tactic": "Credential Access", "severity": "High", "status": "open" }
```

4. Platform Features

4.1 Dashboard Overview

The main dashboard provides a high-level SOC view with four real-time stat cards showing total alerts, critical alerts, high severity alerts, and open (unresolved) alerts fetched live from the Flask API.

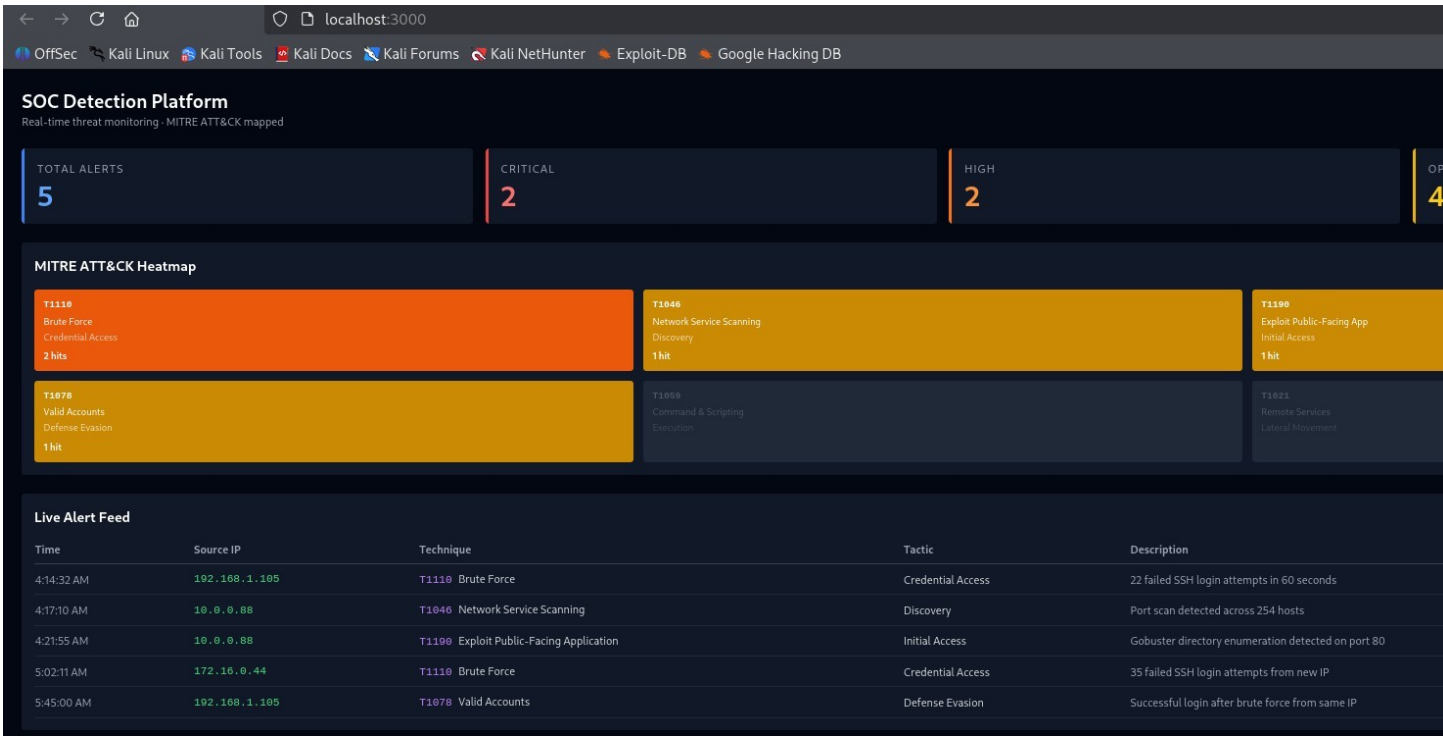

Figure 1: Full SOC Dashboard — Stat Cards, MITRE Heatmap, IP Risk Scores

4.2 MITRE ATT&CK Heatmap & IP Risk Scores

The MITRE ATT&CK heatmap visually maps detected techniques to their corresponding tactics. Color coding indicates hit frequency: red for 3+ hits, orange for 2 hits, yellow for 1 hit, and grey for undetected techniques. The IP Risk Score panel automatically calculates a risk score (0-100) per source IP based on alert severity and frequency, categorizing each as Critical, High, Medium, or Low.

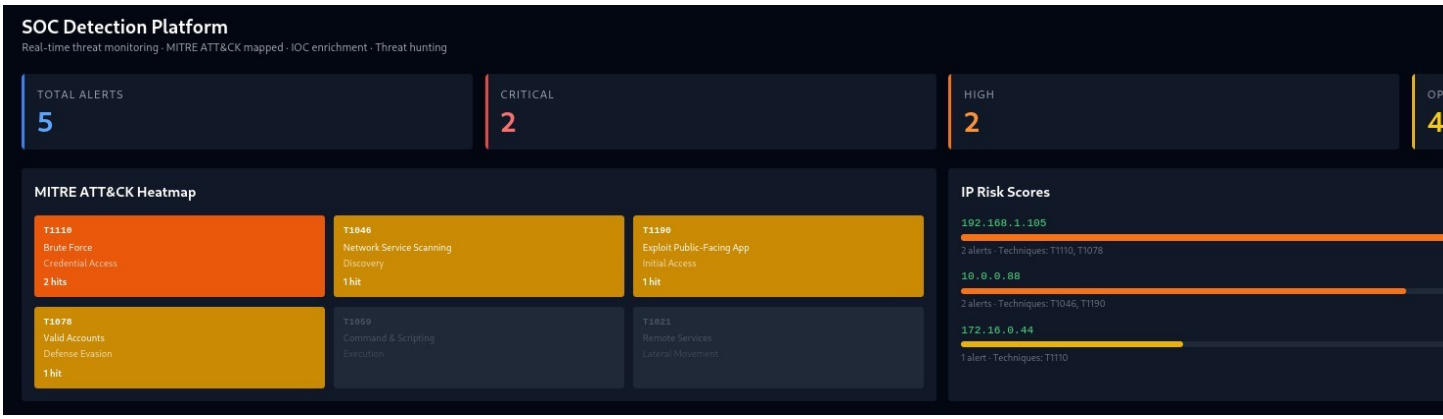

Figure 2: MITRE ATT&CK Heatmap with IP Risk Score Panel

4.3 Threat Hunt Panel

The Threat Hunt Panel provides real-time search and filtering capabilities across all alerts. Analysts can filter by source IP, technique ID (e.g., T1110), or severity level. Results update instantly as the analyst types, mimicking Splunk-style SOC workflows without requiring SPL knowledge.

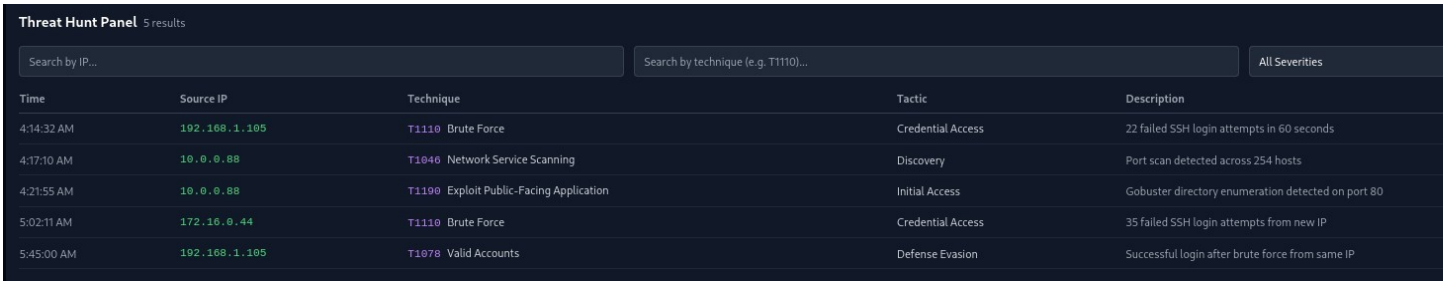

Figure 3: Threat Hunt Panel — Live Search and Filter

4.4 Live Alert Feed

The Live Alert Feed displays all IDS-generated alerts in a structured table with timestamp, source IP, MITRE technique ID and name, tactic, description, and colour-coded severity badge. This mirrors the alert triage workflow used in real SOC environments.

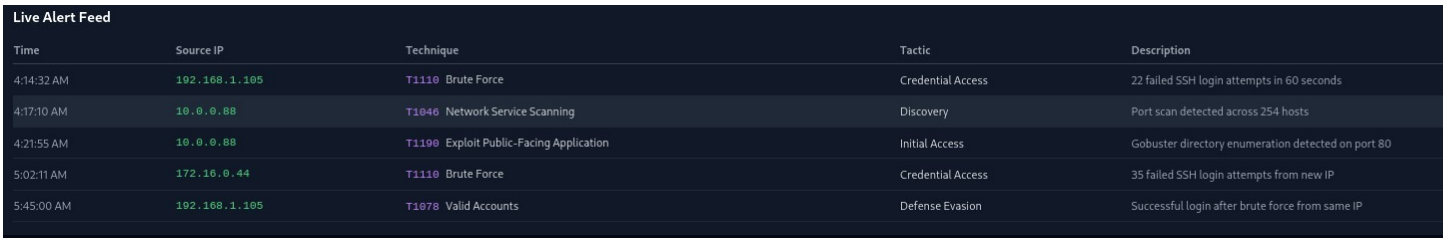

Figure 4: Live Alert Feed with Severity Badges

5. Detection Logic & Risk Scoring

5.1 MITRE ATT&CK Detection Coverage

Technique ID	Technique Name	Tactic	Log Source
T1110	Brute Force	Credential Access	auth.log
T1046	Network Service Scanning	Discovery	syslog
T1190	Exploit Public-Facing App	Initial Access	syslog
T1078	Valid Accounts	Defense Evasion	auth.log

5.2 Risk Scoring Formula

Each source IP is assigned a risk score based on the cumulative severity weight of its alerts, capped at 100:

Severity	Weight	Rationale
Critical	40 points	Immediate threat, active compromise likely
High	25 points	Significant attack activity detected
Medium	10 points	Suspicious but not immediately dangerous
Low	5 points	Informational, low confidence detection

5.3 Key Findings from Demo Dataset

Source IP	Risk Score	Level	Attack Chain
192.168.1.105	65/100	High	Brute Force (T1110) → Valid Accounts (T1078) — full credential compromise chain
10.0.0.88	50/100	High	Port Scan (T1046) → Web Enumeration (T1190) — reconnaissance to exploitation
172.16.0.44	25/100	Medium	Brute Force (T1110) — single technique, early stage

IP 192.168.1.105 represents the most concerning threat actor — it first conducted SSH brute force (T1110) and then successfully logged in with valid credentials (T1078), indicating a complete credential access attack chain that warrants immediate incident response.

6. Setup & Deployment

6.1 Backend Setup

- Navigate to the backend directory: cd backend
- Create a virtual environment: python3 -m venv venv
- Activate it: source venv/bin/activate
- Install dependencies: pip install flask flask-cors requests

- Run the API server: python app.py
- API available at: http://localhost:5000

6.2 Frontend Setup

- Navigate to the frontend directory: cd frontend
- Install dependencies: npm install
- Start the development server: npm start
- Dashboard available at: http://localhost:3000

```
┌──(venv)─(kali㉿kali)-[~/soc-platform/backend]
└─$ source venv/bin/activate
python app.py
 * Serving Flask app 'app'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server f
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 141-140-315
```

Figure 5: Flask API Server Running on Port 5000

7. Development Phases

Phase	Duration	Deliverable
Phase 1	Days 1-3	Flask API with 5 endpoints, React scaffold connected to backend
Phase 2	Days 4-10	Full dashboard UI — stat cards, MITRE heatmap, live alert feed
Phase 3	Days 11-16	IP risk scoring, Threat Hunt Panel, VirusTotal IOC enrichment
Phase 4	Days 17-21	README documentation, GitHub deployment, project report

8. Related Projects

8.1 Log-Based Intrusion Detection System

The IDS that feeds alerts into this platform was built in March 2025 using Python, Bash, and JSON. It parses auth.log and syslog in real time to detect SSH brute-force (T1110), port scanning (T1046), and web enumeration. Alerts are structured as MITRE-mapped JSON objects and were deployed internally at Cloudisian as a blue team simulation exercise.

8.2 Android APK Malware Analysis Sandbox

Built in April 2026, this project creates an automated mobile security pipeline: APK upload → MobSF static/dynamic scan → OWASP Mobile Top 10 enrichment → VirusTotal threat

intelligence lookup → auto-generated PDF pentest report. The VirusTotal API integration in this SOC platform directly extends that experience.

9. Conclusion

This SOC Detection and Threat Hunting Platform demonstrates the ability to build a full analyst workflow from detection through visualization and threat intelligence enrichment. By extending a real IDS with a professional dashboard, the project bridges offensive security knowledge (MITRE ATT&CK, IOC analysis) with practical defensive tooling (alert triage, risk scoring, threat hunting).

The platform is directly applicable to L1/L2 SOC analyst roles where rapid alert triage, MITRE mapping, and IP enrichment are daily tasks. Future enhancements could include real-time log ingestion via WebSockets, Splunk integration for production SIEM data, and automated incident response playbook triggers.